

E-Commerce Sales & Customer Behavior Analysis

Exploratory Data Analysis of the UCI Online Retail dataset using Python

Problem statement. An online retail company wants to understand its sales performance, customer purchasing behaviour, product demand and geographical distribution of customers.

Dataset. UCI Online Retail: 541,909 transactions and 8 columns, from 1 Dec 2010 to 9 Dec 2011. **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn.

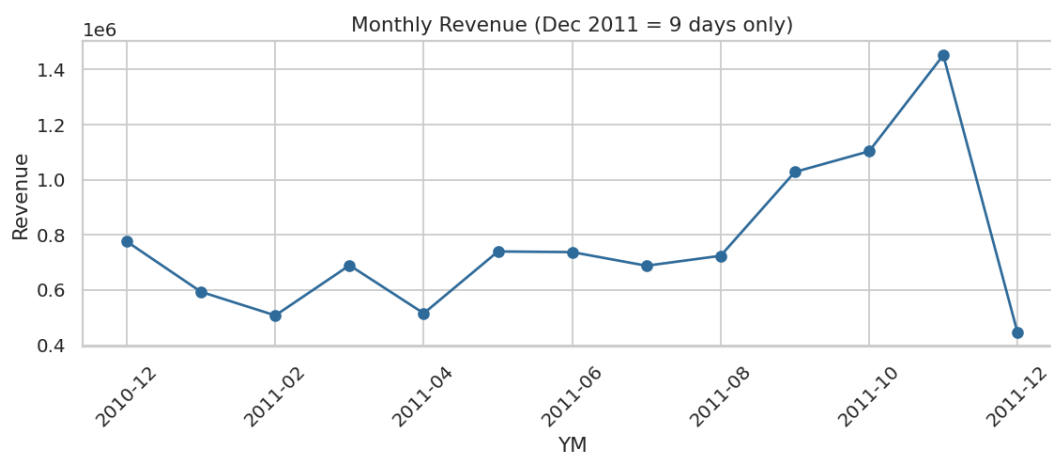
1. Data Understanding and Cleaning

Problems found and the decision taken for each:

Problem	Count	Decision
Exact duplicate rows	5,268	Removed
Cancelled invoices (InvoiceNo starts with C)	9,288 rows	Kept in a separate table (return analysis)
Zero/negative price or quantity	2,517 / 10,624	Removed (bad entries, adjustments)
Non-product entries (postage, fees)	StockCode not 5 digits	Removed
Missing Description	1,454	Removed
Missing CustomerID	135,080 (about 25%)	Rows kept; customer analysis uses only rows with an ID
Invoices 541431 and 581483 (74,215 and 80,995 units)		Removed: both were cancelled later (about 245,000 revenue)

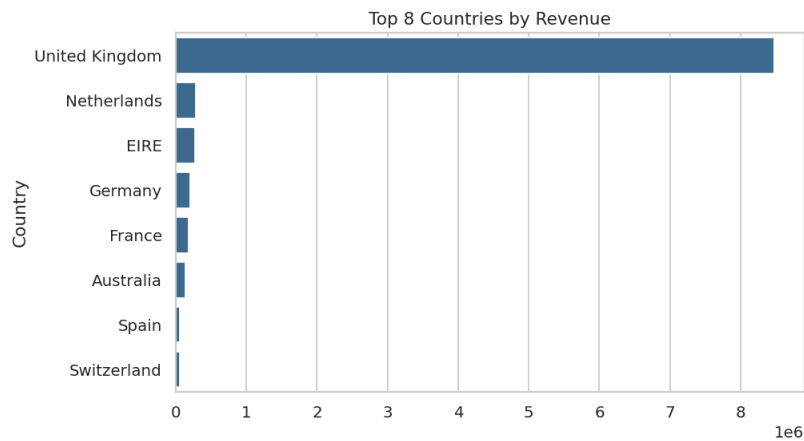
Final dataset: 522,502 rows, 4,333 customers (with ID), 19,771 invoices, 3,899 products, 38 countries, total revenue 10,001,168. Large orders other than the two cancelled ones were kept as genuine wholesale sales.

2. Sales Trend (time series)

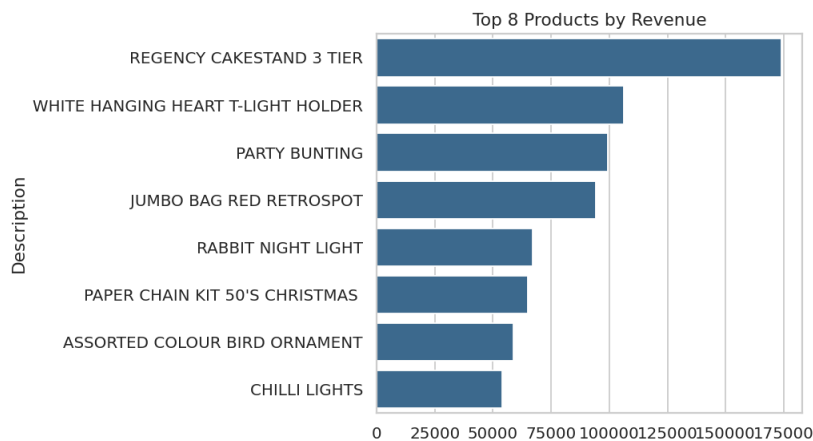


Monthly revenue stays at about 5 to 7.7 lakh from January to August 2011 and then rises to a peak in **November 2011 (1,452,113)**, about 2.9 times the weakest full month (February 2011, 507,783). December 2011 is low only because the data ends on 9 December.

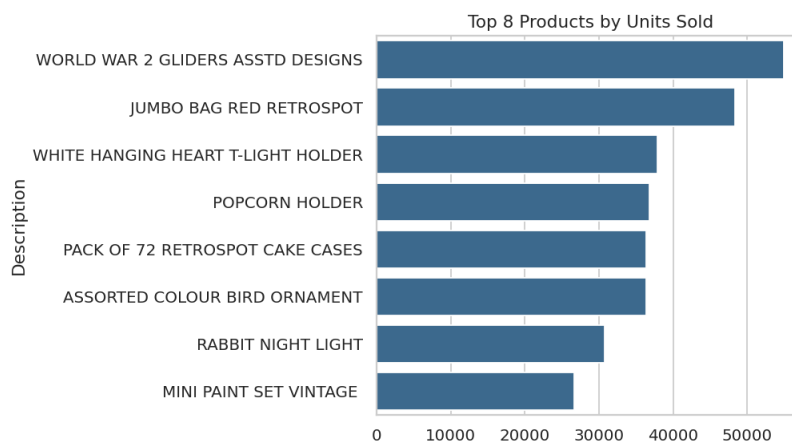
3. Geography, Products and Customers



The **United Kingdom** gives **84.8%** of revenue; the other 37 countries together give 15.2%.

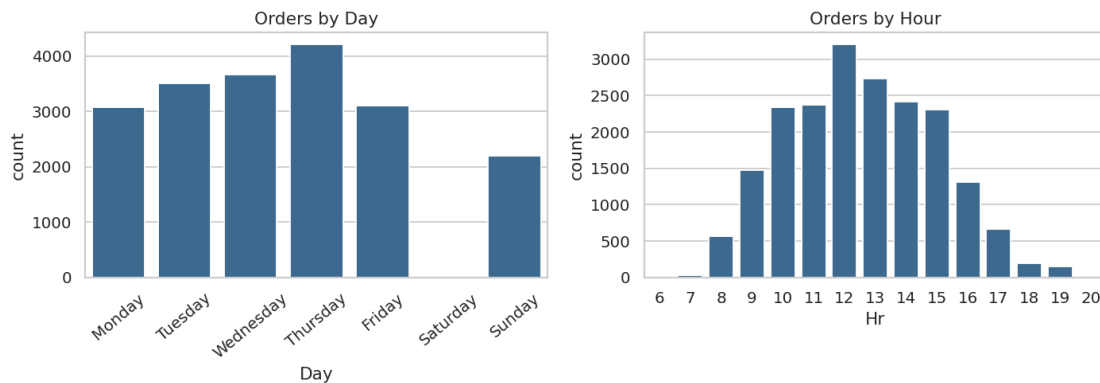


Regency Cakestand 3 Tier is the top product by revenue.



By units, World War 2 Gliders (54,951), Jumbo Bag Red Retrosport (48,371) and White Hanging Heart T-Light Holder (37,872) lead. Top products by revenue and by units are different.

4. Ordering Patterns and Relationships



Thursday is the busiest day (4,209 orders), Sunday the weakest (2,203) and there are no Saturday orders. Orders peak at 12:00 (3,204), so most buying happens in office hours.



Customer spend is strongly linked to items bought (correlation 0.92) and more weakly to the number of orders (0.58). The top 20% of customers give 73.9% of revenue (close to the 80/20 rule). Bulk orders are mostly for cheap products, and expensive items sell in small quantities.

Other findings: average order value is 505.85 but the median is 302.20 (a few very large orders pull the average up). 14.81% of invoices are cancelled (3,836 of 25,900).

5. Product Demand Trend

Comparing the first six months (Dec to May) with the last six months (Jun to Nov), the demand for **White Hanging Heart T-Light Holder (-24.7%)**, Pack of 72 Retrospot Cake Cases (-20.9%) and Assorted Colours Silk Fan (-19.4%) is falling, while **Jumbo Bag Pink Polkadot (+79.6%)** and Red Harmonica in Box (+90.2%) are growing. Popcorn Holder and Rabbit Night Light look like late launches, so their growth percentages are not meaningful.

6. Business Recommendations

- Prepare stock and marketing for September to November, when demand rises most.
- Reduce dependence on the UK with campaigns and shipping offers in the Netherlands, EIRE, Germany and France.
- Give loyalty offers to the top customers, since they bring most of the revenue.

- Investigate the high cancellation rate (stock, shipping or entry errors).
- Send promotions Tuesday to Thursday during office hours.
- Promote or bundle products with falling demand and keep stock of rising products.

7. Limitations

- Data covers only one year, so seasonality cannot be confirmed across years.
- About 25% of rows have no CustomerID, so customer insights cover only the remaining customers.
- The festive-season reason for the November peak is an assumption, not proved by the data.
- There are no Saturday orders in the dataset, which may be a data collection issue.